

insert a battery of V_{γ}
mind the terminals of diode. $V_{\gamma} = 0.3 \text{ V}$ for Ge.
 $V_{\gamma} = 0.3 \text{ V}$ for Ge.

⇒ Applications of Diode

Clippers

Clippers are the diode circuits which are used to clip-off either the complete half-cycle or portion of it.

Clippers

On the basis of position of D & RL

Series

Parallel

On the basis of clipping of half cycle

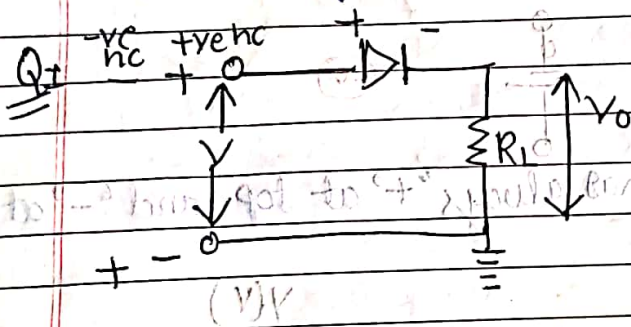
Positive

Negative

⇒ Series Clippers

⇒ Rules for solving clipper problems

- 1) Calculate transition voltage from the circuit. The volt-ages apart from supplied voltages are known as transition voltages.
- 2) Calculate the output during positive half cycle.
- 3) Calculate the output during -ve half cycle.
- 4) Always plot the output waveforms just below the input waveforms.

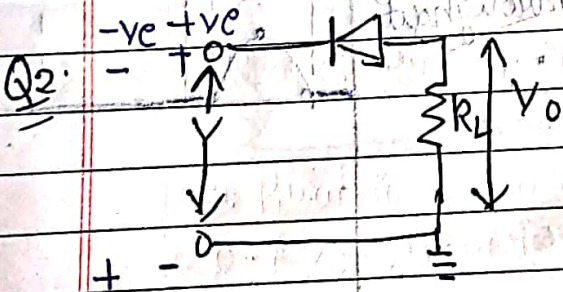
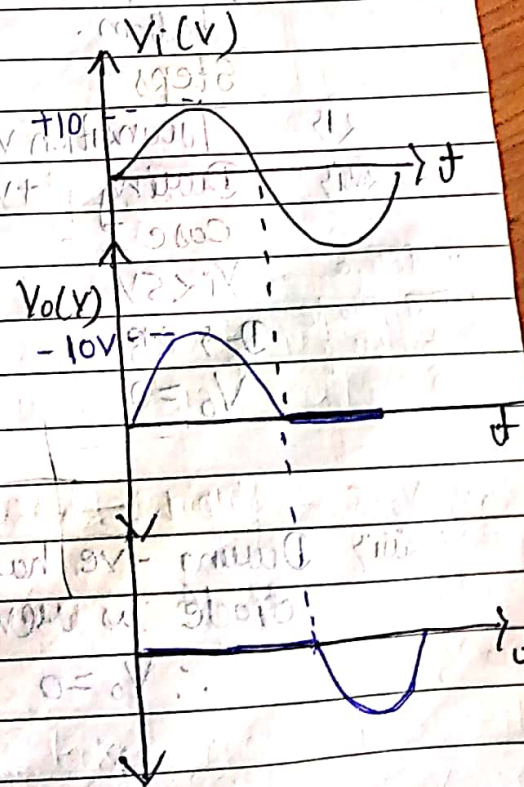


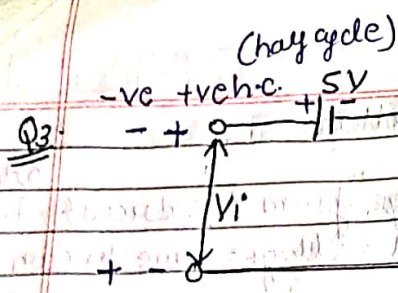
Step-1 Transition voltage = 0

Step 2 During +ve h.c. Diode → FB

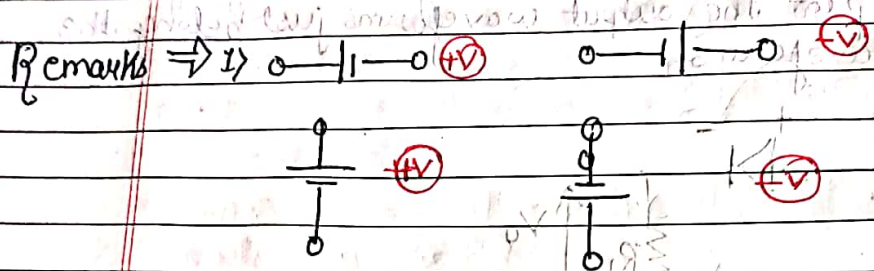
Step 3 During -ve h.c. Diode → RB

$V_0 = 0$





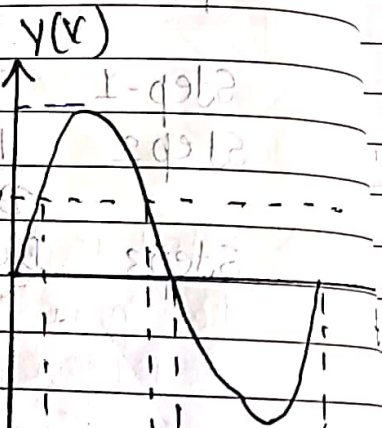
Diode with additional voltage than input voltage are known as **'Biased Diode'**.



2) Output voltage has always '+' at top and '-' at the bottom.

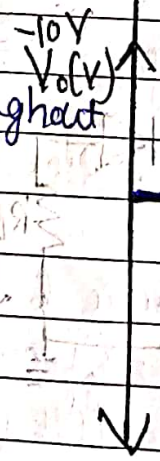
Steps

Transition voltage = 5V	+10V
During +ve half cycle	
Case I	Case II
$V_i < 5V$	$V_i > 5V$
$D \rightarrow RB$	$D \rightarrow FB$
$V_o = 0$	$V_i - 5 - V_o = 0$
	$V_i - V_o = 5$



In $V_i - V_o = 5$
put V_i values
5, 6, 7, 8, 9, 10
6, 7, 8, 9, 10
get V_o

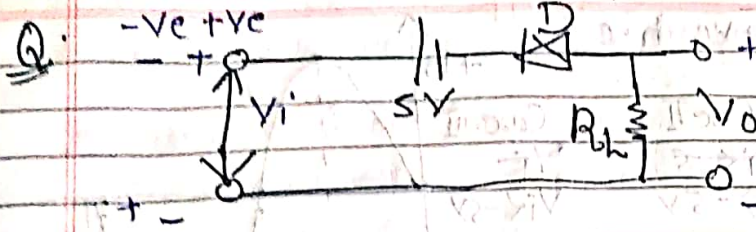
During -ve half cycle, throughout diode is reversed bias.
 $\therefore V_o = 0$



4 combinations :-



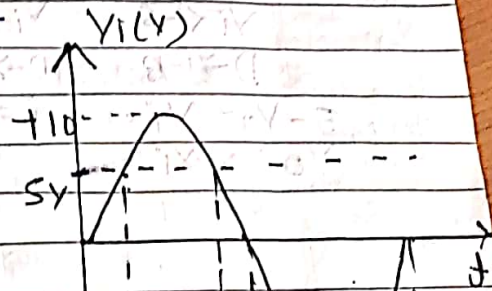
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Step 1: Transition voltage = +5V

Step 2 → During +ve h.c

Case I	Case II	Case III
$V_i < 5$	$V_i > 5$	$V_i < 5$
D → FB	D → RB	D → FB
$V_i - 5 - V_o = 0$	$V_o = 0$	$V_o = V_i - 5$
$V_o = V_i - 5$		



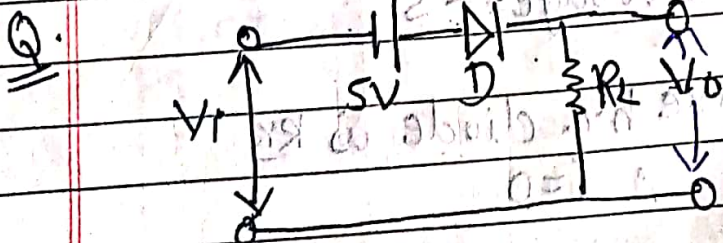
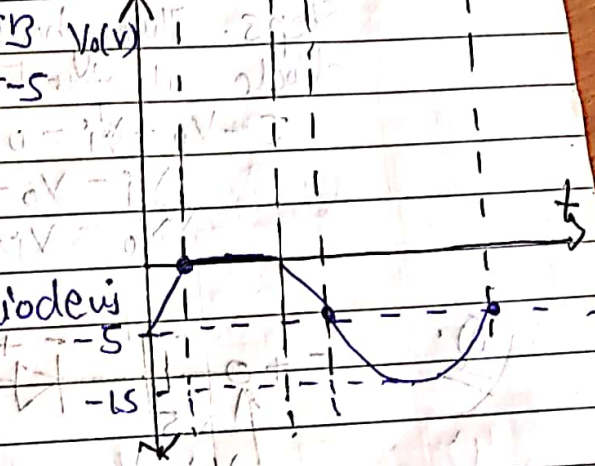
Step-3 During -ve h.c
Through -ve half cycle diode is forward biased

$$-5 - V_o - V_i = 0$$

$$V_o = -V_i - 5$$

$$-10 - 5 = -15$$

-10 sign put + bcz already we are -ve h.c by put V_i sign + forward



Step-1 Transition voltage = +5

Don't use number line concept

Step-2 In +ve h.c

cases where
banege
time
direction
voltage
if we don't
like sign
then in
very easy
quantity
to find

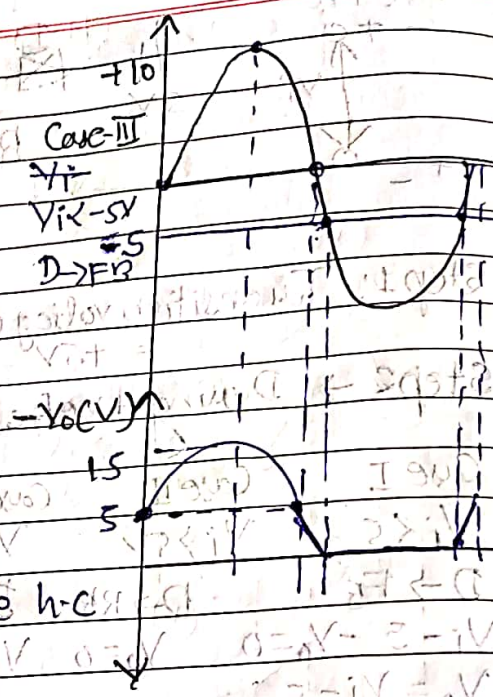
Case I
 $V_i > -5$
 $V_i < -5$
 $D \rightarrow FB$

Case II
 $V_i < -5$
 $V_i > -5V$
 $D \rightarrow RB$

Case III
 V_i
 $V_i < -5V$
 $D \rightarrow FB$

$5 - V_o - V_i = 0$
 $V_o = 5 - V_i$

$V_i(V)$

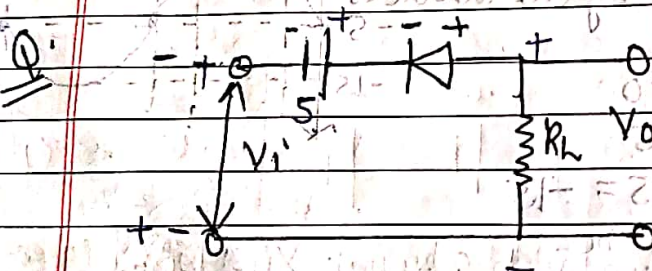


Step 3: Throughout +ve h.c
diode is in FB

$5 - V_o + V_i = 0$

$V_i = V_o - 5$

$V_o = V_i + 5$



Step 1: Transition voltage = -5

Step 2: During +ve h.c diode is in FB
 $i = 0$

Step 3: During -ve h.c

Case-I
 $V_i < -5$
 $D \rightarrow RB$
 $i = 0$

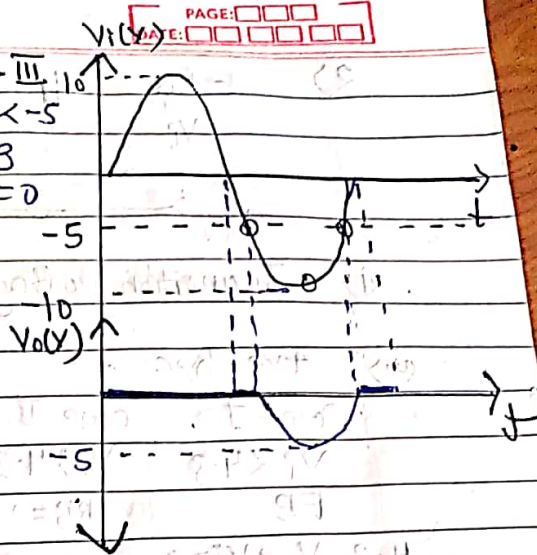
For

Step 1: Tu

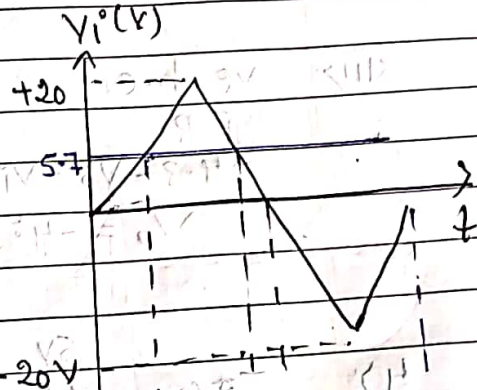
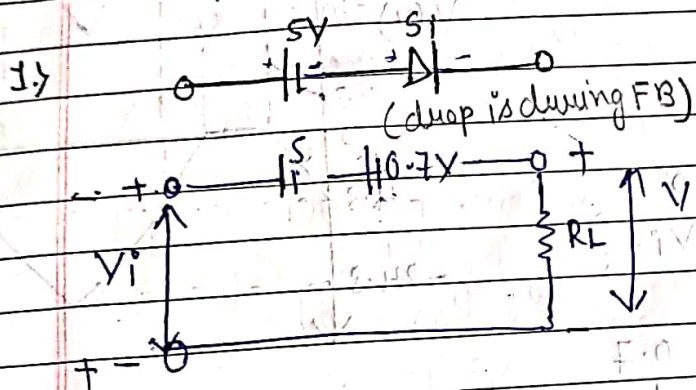
Step 2: D

Step 3: D

Case-I	Case-II	Case-III
$V_i < -5$	$V_i > -5$	$V_i < -5$
$D \rightarrow RB$	$D \rightarrow FB$	RB
$i = 0$	$5 - V_o - V_i = 0$	$i = 0$
	$V_o = 5 - V_i$	



For practical diode



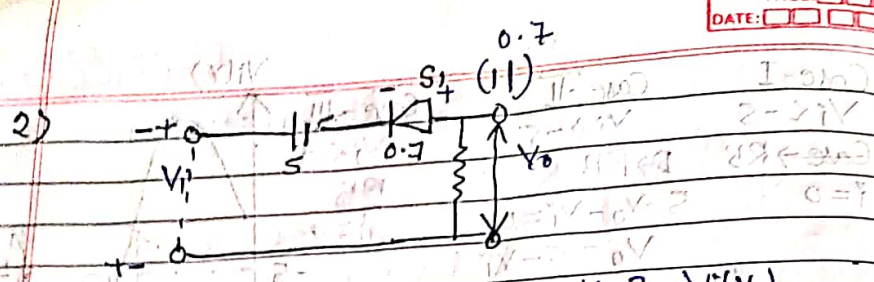
Step 1: Transition voltage = 5.7

Step 2: During +ve half cycle

Case-I	Case-II	Case-III
$V_i < 5.7$	$V_i > 5.7$	$V_i < 5.7$
RB	$-5.7 - V_o + V_i = 0$	RB
$i = 0$	$V_o = V_i - 5.7$	$i = 0$



Step 3: During -ve half cycle
Reverse bias



i) Transition voltage $5 - 0.7 = 4.3$ $V_i(V)$

ii) +ve h.c

Case-I

$V_i < 4.3$
FB

$$-4.3 - V_o + V_i = 0$$

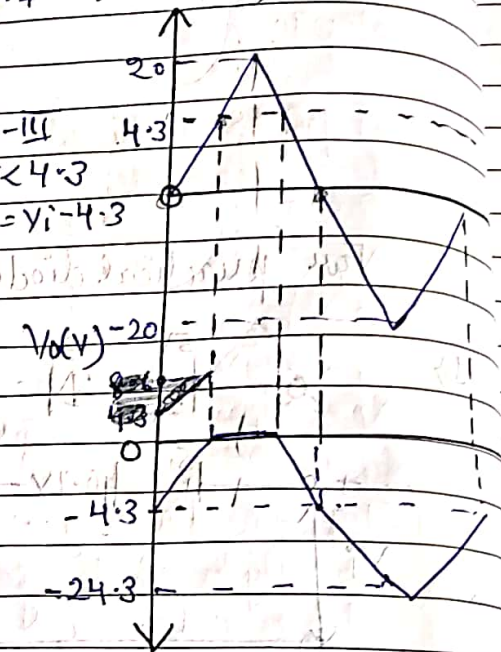
$$V_o = V_i - 4.3$$

Case-II

$V_i > 4.3$
RB=0

Case-III

$V_i < 4.3$
 $V_o = V_i - 4.3$

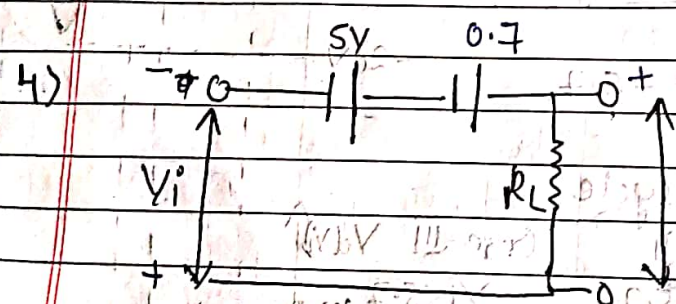


iii) -ve h.c

FB

$$-4.3 - V_o - V_i = 0$$

$$V_o = -4.3 - V_i$$



i) Transition voltage - 5.7

ii) -ve h.c

Case i)

$V_i < -5.7$
RB
 $i=0$

Case ii)

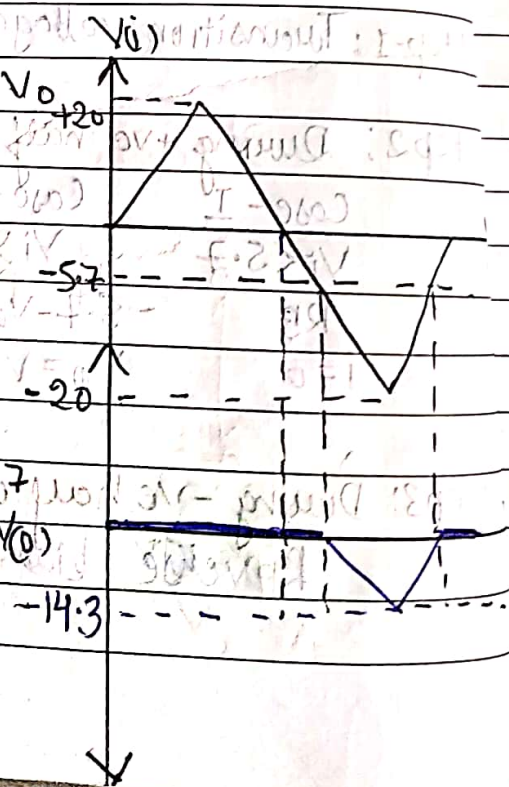
$V_i > -5.7$
RB
 $i=0$

Case iii)

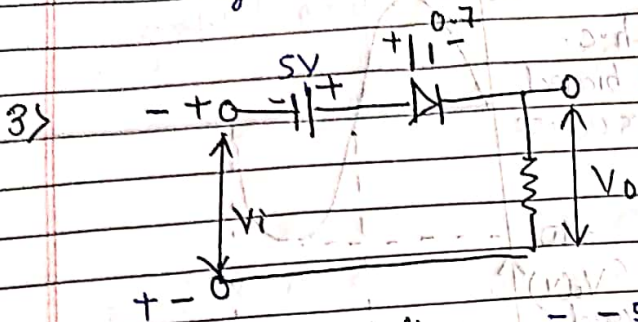
$V_i < -5.7$
 $i=0$

$$5.7 - V_o - V_i = 0$$

$$V_o = 5.7 - V_i$$



iii) +ve h-c
always R.B



Transition voltage = $-5 + 0.7 = -4.3$

iii) +ve half cycle FB

$$5 - 0.7 - V_o + V_i = 0$$

$$V_o = V_i + 4.3$$

iii) -ve half cycle

Case I
 $V_i < -4.3$
FB

Case II
 $V_i > -4.3$
RB

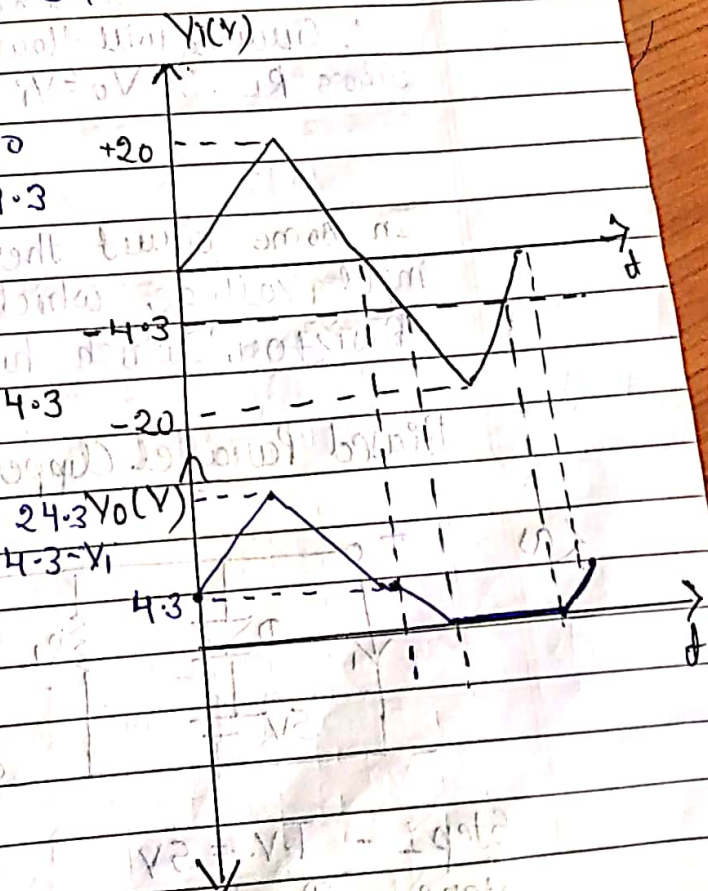
Case III

$V_i < -4.3$
FB

$$5 - 0.7 - V_o - V_i = 0$$

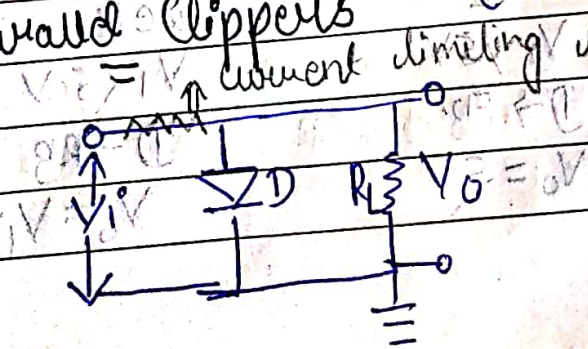
$$V_o = 4.3 - V_i$$

$$V_o = 4.3 - V_i$$



⇒ Parallel Clippers

(Use concept // me branch me voltages same hata hai)



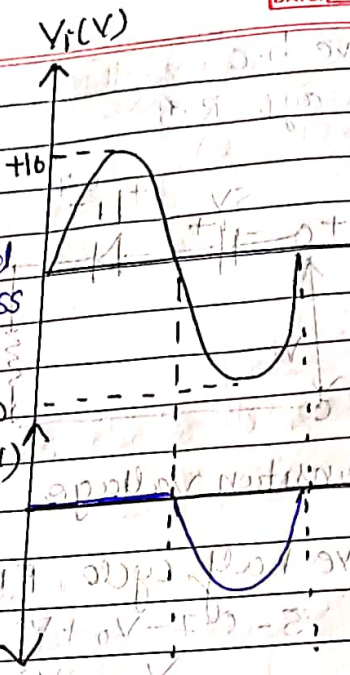
Step-1

Transition Voltage = 0

Step-2 In the +ve h.c.
Diode is forward biased
 $\therefore$ no current flows across
 $R_L \therefore V_o = 0$

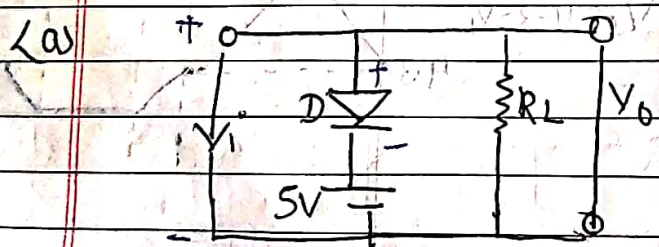
Step-3

In the -ve h.c. $V_o(V)$
Diode is Reverse biased
 $\therefore$ Current will flow
across $R_L \therefore V_o = V_i$



In some circuit there is resistance attached to input voltage which is known as 'Current limiting Resistor' which hinders flow of current. Just keep

Biased Parallel Clippers



Step 1 - T.V. = 5V

Step 2 - During +ve h.c

Case I

$V_i < 5V$

$D \rightarrow RB$

$V_o = V_i$

Case II

$V_i > 5V$

$D \rightarrow FB$

$V_o = 5V$

Case III

$V_i < 5V$

$D \rightarrow RB$

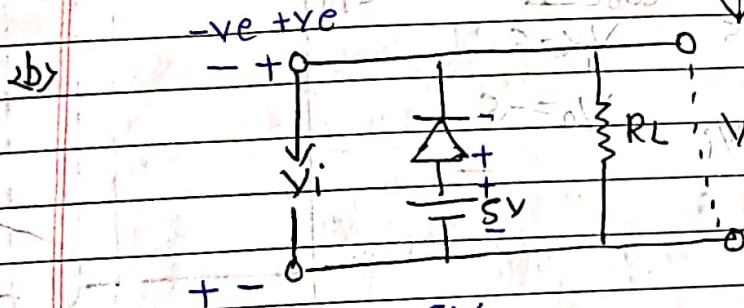
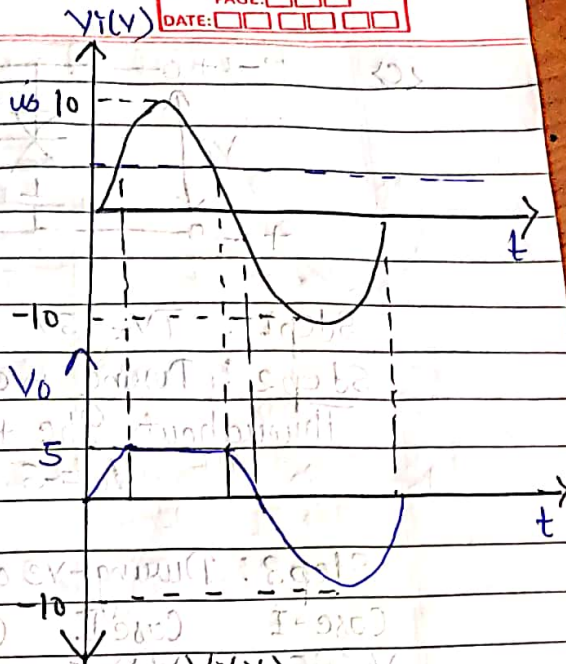
$V_o = V_i$



In // clipper agar FB hai to Load se V_o current nahi Jayega

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Step 3 - During -ve h.c
Throughout the -ve h.c diode is reverse biased.



Step 1 - $T.V. = 5V$

Step 2 - During +ve h.c

Case I

$$V_i < 5V$$

$D \rightarrow FB$

$$\therefore V_o = 5$$

Case II

$$V_i > 5V$$

$D \rightarrow RB$

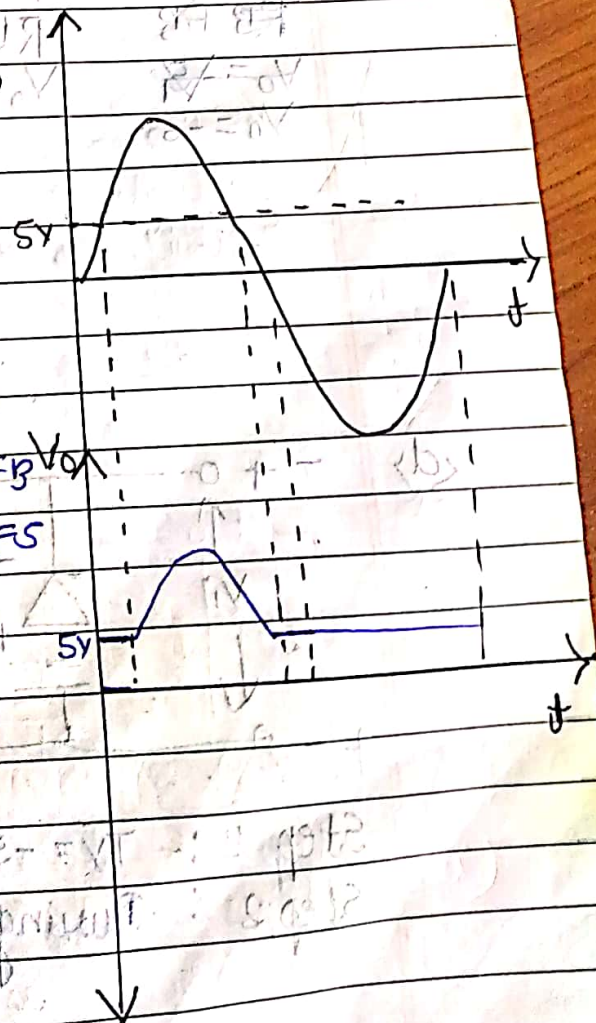
$$V_o = V_i$$

Case III

$$V_i < 5$$

$D \rightarrow FB$

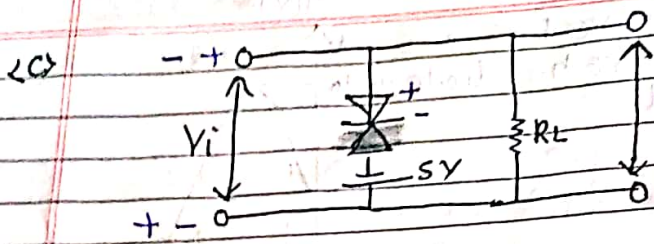
$$\therefore V_o = 5$$



Step 3 - During -ve h.c

$D \rightarrow FB$

$$V_o = 5$$



Step 1: $V_T = -5V$

Step 2: During +ve h.c
Throughout the +ve h.c diode is FB
 $V_O = -5$

Step 3: During -ve cycle

Case-I

Case-II

Case-III

$V_i < -5$

$V_i > -5$

$V_i < -5$

FB

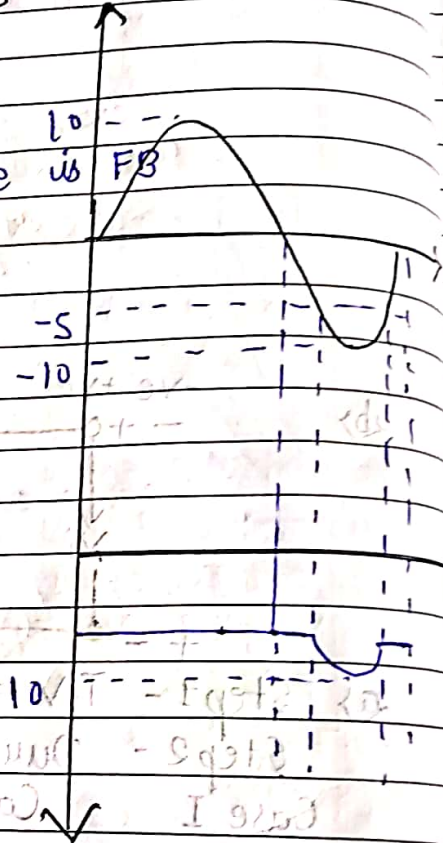
FB

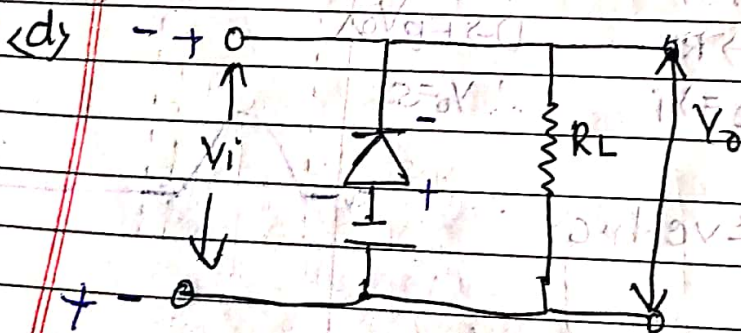
$V_O = -5$

$V_O = -5$

$V_O = V_i$

$V_O = -5$

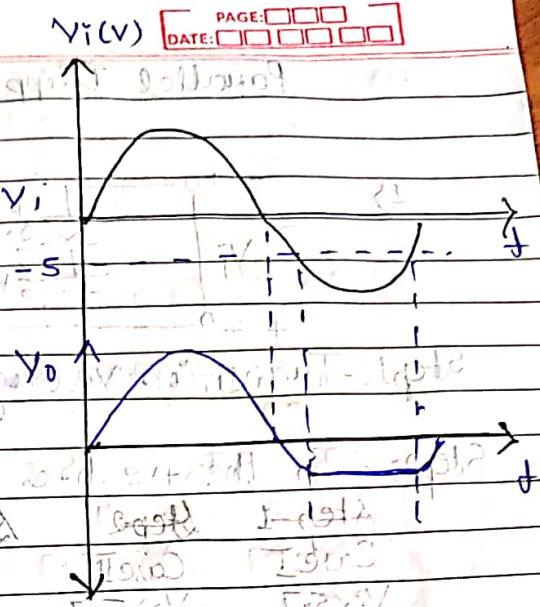




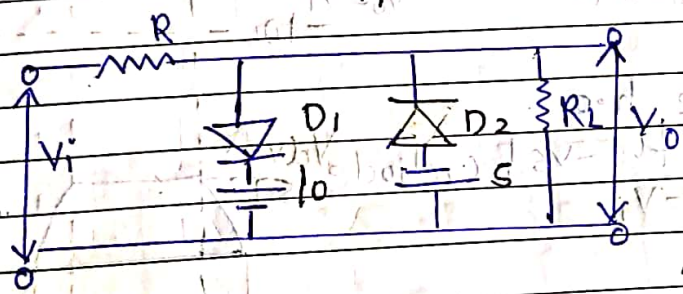
Step 1: $V_T = -5$

Step 2: During +ve h.c, RB $\therefore V_O = V_i$

Step 3: Drawing -ve h.c.
 Case I Case II Case III
 $V_i < -5$ $V_i > -5$ $V_i < -5$
 RB FB $V_o = V_i$
 $V_o = V_i$ $\therefore V_o = -5$

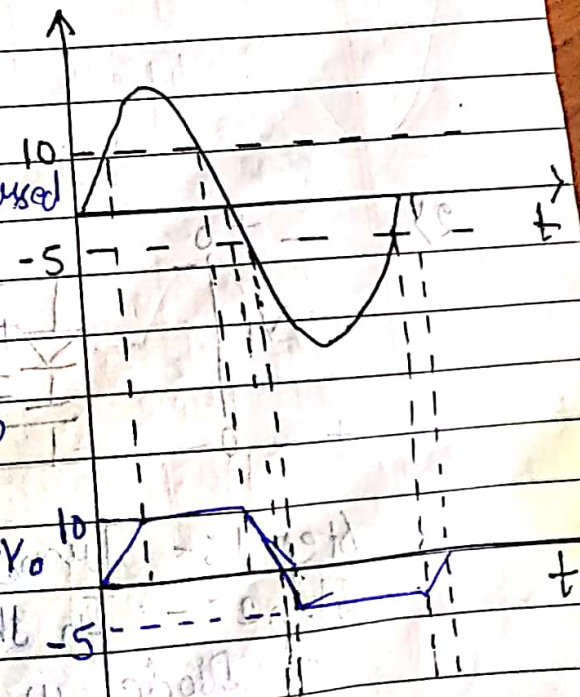


 ⇒ Double Ended Clipper / Combination clipper



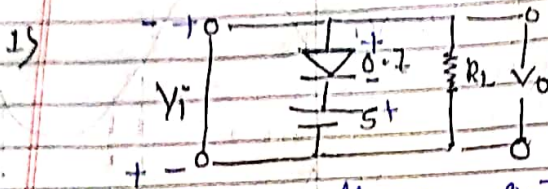
Step 1: $V_i = +10, -5V$
 Step 2: In +ve h.c
 Throughout +ve h.c. D_2 is reversed bias

$D_1 \Rightarrow FB$
 Case I Case II Case III
 $V_i < 10$ $V_i > 10$ $V_i < 10$
 RB $V_o = 10$ RB
 $V_i = V_o$ $V_i = V_o$



Step 2: In -ve h.c
 Throughout -ve h.c. D_1 is RV
 Case I Case II Case III
 $V_i < -5$ $V_i > -5$ $V_i < -5$
 RB FB RB
 $V_i = V_o$ $V_o = -5$ $V_i = V_o$

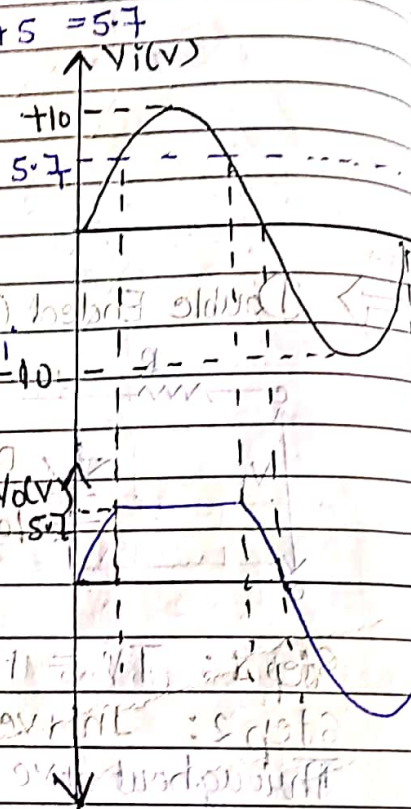
Parallel Clipper with practical Diodes



Step 1:- Transition voltage $0.7 + 5 = 5.7$

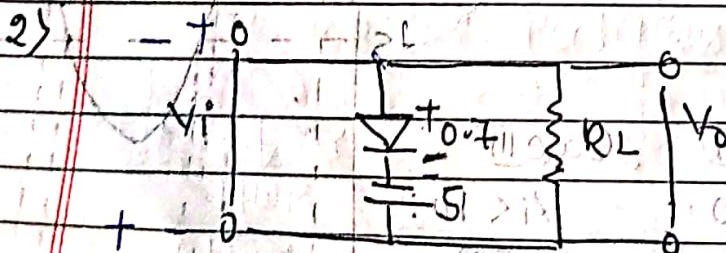
Step 2:- In the +ve h.c.

Case I	Case II	Case III
$V_i < 5.7$	$V_i > 5.7$	$V_i < 5.7$
RB	FB	RB
$V_o = V_i$	$V_o = 5.7$	$V_o = V_i$



Step 3: In the -ve h.c.
Throughout the -ve h.c, diode is RB $V_o = V_i$

3)



Step 1:- Transition voltage $0.7 - 5 = -4.3$

Step 2:- In the +ve h.c. Throughout the +ve h.c. Diode is FB $V_o = 0$

$V_o = 0$